

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

November – December 2011

CL 201 Surveying

Date: 29.11.2011, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Describe briefly the uses of various accessories of plane table [05]
(b) Define Hydrography. What do you understand by 'Sounding'? [02]
- Q - 2 (a) Write a short note on Profile leveling. [06]
(b) Explain step by step procedure of measuring deflection angle with the help of a theodolite. [04]
(c) Define the following operations of theodolite: telescope normal, transiting, swinging, permanent adjustment [04]

OR

- Q - 2 (a) 1. What are the different methods of reduction of level? Which of them you prefer? [06]
2. Show characteristics features of the contour lines for: pond, hill, valley, uniformly sloping ground, vertical cliff.
- (b) Enlist and explain different checks for a close traverse. [04]
(c) Estimate the closing error for traverse ABCDE given below and correct it with the help of Bowditch's rule. [04]

Line	Length	Bearing
AB	105.8	N 40° 45' W
BC	142.5	N 51° 30' E
CD	188.8	N 48° 15' E
DA	188.9	S 76° 45' W

- Q - 3 (a) Explain with neat sketch the method of Resection of plane table surveying. Discuss the application of method of resection. [07]
(b) Describe briefly temporary adjustments of a theodolite [04]
(c) Enlist various methods of locating sounding and explain any one of them in brief. [03]

OR

- Q - 3 (a) What is two point problem? How it is solved? [07]

- (b) Explain with neat sketch how main scale and vernier scale of a transit theodolite are graduated to obtain least count of 20". [04]
- (c) What is unit of sounding? Explain the use of Fathometer. [03]

SECTION – II

- Q - 4 (a)** Describe the method of setting out a simple circular curve by Rankine's deflection angle method [04]
- (b) State trapezoidal rule. What are its limitations? [03]
- Q - 5 (a)** Write a short note on Total station. [06]
- (b) Calculate the ordinates at every 5 m distance for a circular curve having a long chord of 40 m and versed sine of 2 m. [05]
- (c) What is vertical curve? Why it is provided? [03]

OR

- Q - 5 (a)** Explain the use of Electronic theodolite in brief. [06]
- (b) Write a short note on Superelevation [05]
- (c) Draw neat sketch of a simple circular curve with proper notations and explain the term 'degree of curve'. [03]
- Q - 6 (a)** Explain in brief the steps of locating pier of a bridge. [06]
- (b) What is Simpson's rule? Derive expression for it. [04]
- (c) Derive formula for area of a two level section. [04]

OR

- Q - 6 (a)** Write a brief note on prismoidal correction. [06]
- The table below gives ground level along centre line of an 8m wide road with side slope 1:1 proposed on it. Considering constant formation level at 115m calculate the total volume of earthwork required using: 1) Prismoidal formula and 2) Trapezoidal formula.

Chainage (m)	0	50	100	150	200	250	300
GL (m)	116.5	115.4	117.5	118.2	116.3	115.6	117.1

- (b) What do you understand by setting out works? How it differs from other surveying practices? [06]
- (c) Draw neat sketch of a multilevel section and explain method of finding area of such section? [02]
